Solution of Tutorial 7: Laplace Transform

Formulas on Laplace transform:

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}, \quad s > 0, \quad \mathcal{L}[e^{at}] = \frac{1}{s-a}, \quad s > a,$$
$$\mathcal{L}[\cos bt] = \frac{s}{s^2+b^2}, \quad s > 0,$$
$$\mathcal{L}[\sin bt] = \frac{b}{s^2+b^2}, \quad s > 0, \quad \mathcal{L}[t^{-1/2}] = \sqrt{\frac{\pi}{s}}, \quad s > 0,$$
$$\mathcal{L}[e^{at} f] = F(s-a), \quad \mathcal{L}\big[u_a(t) f(t-a)\big] = e^{-as} F(s), \quad \text{where } \mathcal{L}[f(t)] = F(s),$$
$$\mathcal{L}[\delta(t-t_0)] = e^{-st_0}, \quad t_0 > 0; \qquad \mathcal{L}[\delta(t)] = 1,$$
$$\mathcal{L}[f'(t)] = s\mathcal{L}[f(t)] - f(0), \quad \mathcal{L}[f''(t)] = s^2\mathcal{L}[f(t)] - sf(0) - f'(0).$$

1. Find the Laplace transforms of

$$f(t) = 4e^{3t} + 2\sin(5t) - 7t^3.$$

Solution: By the linearity of Laplace transform, we have

$$\begin{aligned}\mathcal{L}[f(t)] &= \mathcal{L}[4e^{3t} + 2\sin(5t) - 7t^3] = 4\mathcal{L}[e^{3t}] + 2\mathcal{L}[\sin(5t)] - 7\mathcal{L}[t^3] \\ &= \frac{4}{s-3} + 2\frac{5}{s^2+5^2} - 7\frac{3!}{s^4} = \frac{4}{s-3} + \frac{10}{s^2+25} - \frac{42}{s^4}, \quad s > 3.\end{aligned}$$

2. Show that the Laplace transform of $\sqrt{t}$ is $\sqrt{\pi}/(2s^{3/2})$ with $s > 0$.

Solution: By the definition of Laplace transform and integration by parts, we find

$$\begin{aligned}\mathcal{L}[\sqrt{t}] &= \int_0^\infty e^{-st}\sqrt{t}\,dt = -\frac{1}{s}\int_0^\infty \sqrt{t}\,d\,e^{-st} \\ &= -\frac{1}{s}\sqrt{t}e^{-st}\Big|_{t=0}^\infty + \frac{1}{2s}\int_0^\infty \frac{1}{\sqrt{t}}e^{-st}dt \\ &= 0 + \frac{1}{2s}\mathcal{L}[1/\sqrt{t}] = \sqrt{\pi}/(2s^{3/2}).\end{aligned}$$

Another approach: Note that

$$(\sqrt{t})' = \frac{1}{2\sqrt{t}}$$

and recall the derivative formula

$$\mathcal{L}[f'] = s\mathcal{L}[f] - f(0).$$

Let $f = \sqrt{t}$. We find

$$\mathcal{L}[(\sqrt{t})'] = s\mathcal{L}[\sqrt{t}] - 0 \quad \Rightarrow \quad \mathcal{L}[\sqrt{t}] = \frac{1}{s}\mathcal{L}[(\sqrt{t})'] = \frac{1}{s}\mathcal{L}\Big[\frac{1}{2\sqrt{t}}\Big] = \sqrt{\pi}/(2s^{3/2}).$$

3. Show that if $f(t)$ is piecewise continuous and of exponential order on $(0, +\infty)$, then its Laplace transform $F(s) \to 0$ as $s \to +\infty$.

Proof: Suppose that $f(t)$ has n pieces with jump continuities at $t = b_i$, $1 \le i \le n$ with $0 = b_0 < b_1 < b_2 < \cdots < b_n < \infty$. As $f(t)$ is of exponential order, then there exist $M > 0$ and $\alpha > 0$ such that

$$|f(t)| \le Me^{\alpha t}, \quad t > T.$$

We next show that this bound holds for all $t > 0$. So we consider $f(t)$ for $t \in [0, T]$. If $b_n < T$, then for each piece, say on $[b_{i-1}, b_i]$, $f(t)$ is continuous by defining $f(b_{i-1}) = f(b_{i-1}+)$, and $f(b_i) = f(b_i-)$, so

$$|f(t)| \le M_i = \max_{t \in [b_{i-1}, b_i]} |f(t)|, \quad 1 \le i \le n.$$

Then we take $\tilde{M} = \max\{M, M_1, \cdots, M_n\}$, we obtain

$$|f(t)| \le \tilde{M}e^{\alpha t}, \quad t \ge 0,$$

which is obvious true if $T \le b_n$.

We next show that $\lim\limits_{s \to \infty} F(s) = 0$. We have

$$\begin{aligned} |F(s)| = |\mathcal{L}[f(t)]| &= \Big| \int_0^\infty f(t)e^{-st} dt \Big| \le \int_0^\infty |f(t)|e^{-st} dt \\ &\le \int_0^\infty \tilde{M}e^{(\alpha - s)t} dt = \frac{\tilde{M}}{s - \alpha}, \quad s > \alpha. \end{aligned}$$

Thus, we can claim $\lim_{s\to\infty} F(s) = 0$ by the Sandwich theorem in limits.

4. Let

$$f(t) = \begin{cases} (\sin t)/t, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

Find the Taylor series for f about $t = 0$. Assuming that the Laplace transform of this function can be computed term by term, show that

$$\mathcal{L}[f] = \arctan\Big(\frac{1}{s}\Big), \quad s > 0.$$

Proof: Using the Taylor's series of $\sin t$, we find

$$\begin{aligned} f(t) &= \frac{\sin t}{t} = \frac{1}{t}\Big(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots\Big) \\ &= 1 - \frac{t^2}{3!} + \frac{t^4}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n+1)!}. \end{aligned}$$

Thus we have

$$\begin{aligned} F(s) = \mathcal{L}[f(t)] &= \int_0^\infty f(t)e^{-st}dt = \sum_{n=0}^{\infty} \int_0^\infty \frac{(-1)^n t^{2n}}{(2n+1)!} e^{-st} dt \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \mathcal{L}[t^{2n}] = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{(2n)!}{s^{2n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{1}{s^{2n+1}} \\ &= \arctan\Big(\frac{1}{s}\Big), \end{aligned}$$

where we used the Taylor expansion of $\arctan x$.

5. Find the inverse Laplace transform of

$$\text{(i)} \quad \frac{s+4}{s^2+6s+13}, \qquad \text{(ii)} \quad \frac{2e^{-s}}{s^2+4}, \qquad \text{(iii)} \quad \frac{e^{-5s}}{2s^2+s+2}.$$

Solution: (i) We use the first-shifting theorem: $\mathcal{L}^{-1}[F(s-a)] = e^{at}f(t)$. Then

$$\begin{aligned} \frac{s+4}{s^2+6s+13} &= \frac{s+4}{(s+3)^2+2^2} = \frac{(s+3)+1}{(s+3)^2+2^2} \\ &= \frac{s+3}{(s+3)^2+2^2} + \frac{1}{2}\frac{2}{(s+3)^2+2^2} \end{aligned}$$

Then

$$\mathcal{L}^{-1}\left[\frac{s+4}{s^2+6s+13}\right] = e^{-3t}\cos(2t) + \frac{1}{2}e^{-3t}\sin(2t).$$

(ii) We use the second-shifting theorem: $\mathcal{L}^{-1}[e^{-as}F(s)] = u_a(t)f(t-a)$. We extract $F(s)$ and work out its inverse Laplace transform $f(t)$:

$$F(s) = \frac{2}{s^2+4} \quad \Rightarrow \quad f(t) = \mathcal{L}^{-1}\left[\frac{2}{s^2+4}\right] = \sin(2t).$$

Thus, by the second-shifting theorem,

$$\mathcal{L}^{-1}\left[\frac{2e^{-s}}{s^2+4}\right] = u_1(t)\sin(2(t-1)).$$

(iii) Similar to (ii), we extract $F(s)$ and work out its inverse Laplace transform $f(t)$:

$$F(s) = \frac{1}{2s^2+s+2} = \frac{1}{2}\frac{1}{(s+\frac{1}{4})^2+\frac{15}{16}} = \frac{1}{2}\frac{4}{\sqrt{15}}\frac{\frac{\sqrt{15}}{4}}{(s+\frac{1}{4})^2+\frac{15}{16}}$$
$$\Rightarrow \quad f(t) = \frac{2}{\sqrt{15}}e^{-\frac{t}{4}}\sin\Big(\frac{\sqrt{15}}{4}t\Big).$$

Thus, by the second-shifting theorem,

$$\mathcal{L}^{-1}\left[\frac{e^{-5s}}{2s^2+s+2}\right] = u_5(t)f(t-5), \qquad f(t) = \frac{2}{\sqrt{15}}e^{-\frac{t}{4}}\sin\Big(\frac{\sqrt{15}}{4}t\Big).$$

6. Use Laplace transform to solve the equation

$$y'' + 2y' + y = e^{-x}; \quad y(1) = y'(1) = 0.$$

Solution: Notice that the initial values are not set at 0, so we first make a change to variable: $t = x - 1$ and convert the equation as

$$y''(t) + 2y'(t) + y(t) = e^{-(t+1)}, \quad y(0) = y'(0) = 0,$$

where $y(x) = y(t)$. Denote $Y(s) = \mathcal{L}[y(t).]$ Apply the Laplace transform to the ODE:

$$(s^2+2s+1)Y(s) = \mathcal{L}[e^{-(t+1)}] = \frac{1}{e}\frac{1}{s+1}.$$

Therefore,

$$Y(s) = \frac{1}{e}\frac{1}{(s+1)^3} \quad \Rightarrow \quad y(t) = \frac{1}{2e}t^2e^{-t} \quad \Rightarrow \quad y(x) = \frac{(x-1)^2}{2}e^{-x}.$$

7. Use Laplace transform to solve the equation

$$y'' + 4y = f(t) = \begin{cases} 0, & 0 \le t < 5, \\ (t-5)/5, & 5 \le t < 10, \\ 1, & t \ge 10, \end{cases}$$

with the initial condition: $y(0) = y'(0) = 0$.

Solution: We first express $f(t)$ in terms of unit step functions:

$$\begin{aligned} f(t) &= 0 \cdot (u_0(t) - u_5(t)) + \frac{t-5}{5}(u_5(t) - u_{10}(t)) + 1 \cdot (u_5(t) - u_\infty(t)) \\ &= (u_5(t)(t-5) - u_{10}(t)(t-10))/5 \end{aligned}$$

By the second-shifting theorem, we have

$$\mathcal{L}[f] = \frac{e^{-5s} - e^{-10s}}{5s^2}.$$

Denote $Y(s) = \mathcal{L}[y]$. Then the Laplace transformed equation becomes

$$\begin{aligned} & (s^2+4)Y(s) = \frac{e^{-5s} - e^{-10s}}{5s^2} \\ \Rightarrow \quad & Y(s) = \frac{e^{-5s}}{5s^2(s^2+4)} - \frac{e^{-10s}}{5s^2(s^2+4)} = e^{-5s}F(s) - e^{-10s}F(s), \end{aligned}$$

where we denote and can find its inverse Laplace transform:

$$F(s) = \frac{1}{5s^2(s^2+4)} = \frac{1}{20}\left(\frac{1}{s^2} - \frac{1}{s^2+4}\right) \quad \Rightarrow \quad h(t) = \mathcal{L}^{-1}[F(s)] = \frac{t}{20} - \frac{1}{40}\sin(2t).$$

Thus, by the second-shifting theorem, we find the solution:

$$y(t) = u_5(t)h(t-5) - u_{10}(t)h(t-10), \quad h(t) = \mathcal{L}^{-1}[F(s)] = \frac{t}{20} - \frac{1}{40}\sin(2t).$$

8. Use Laplace transform to solve the system

$$\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0, \\ x_1' - 2x_1 + x_2' = -2e^{-t}, \end{cases}$$

with the initial condition

$$x_1(0) = 3, \quad x_1'(0) = 2, \quad x_2(0) = 0.$$

Solution: Let $X_1(s) = \mathcal{L}[x_1]$ and $X_2(s) = \mathcal{L}[x_2]$. Apply the Laplace transform to the system:

$$\begin{cases} s^2X_1 - sx_1(0) - x_1'(0) - 2(sX_1 - x_1(0)) - sX_2 + x_2(0) + 2X_2 = 0, \\ sX_1 - x_1(0) - 2X_1 + sX_2 - x_2(0) = -2\mathcal{L}[e^{-t}], \end{cases}$$

that is

$$\begin{cases} s(s-2)X_1 - (s-2)X_2 = 3s - 4, \\ (s-2)X_1 + sX_2 = \dfrac{3s+1}{s+1}. \end{cases}$$

We solve the system by using elimination. We multiply the second equation by s :

$$s(s-2)X_1 + s^2X_2 = \frac{s(3s+1)}{s+1},$$

which together with the first equation implies

$$\begin{aligned} (s^2 + s - 2)X_2 &= \frac{s(3s+1)}{s+1} - (3s-4) \\ \Rightarrow \quad X_2 &= \frac{s(3s+1) - (3s-4)(s+1)}{(s+2)(s-1)(s+1)} = \frac{2}{(s+1)(s-1)}. \end{aligned}$$

With this, we substitute X_2 into the first equation and find

$$\begin{aligned} X_1 &= \frac{1}{s}X_2 + \frac{3s-4}{s(s-2)} = \frac{2}{s(s+1)(s-1)} + \frac{3s-4}{s(s-2)} \\ \Rightarrow \quad X_1 &= \frac{2(s-2) + (3s-4)(s+1)(s-1)}{s(s+1)(s-1)(s-2)} = \frac{3s^2 - 4s - 1}{(s+1)(s-1)(s-2)}. \end{aligned}$$

We have the following partial fractions:

$$\begin{aligned} X_1(s) &= \frac{3s^2 - 4s - 1}{(s+1)(s-1)(s-2)} = \frac{1}{s-1} + \frac{1}{s+1} + \frac{1}{s-2}, \\ X_2(s) &= \frac{2}{(s+1)(s-1)} = \frac{1}{s-1} - \frac{1}{s+1}. \end{aligned}$$

Find the inverse Laplace transforms:

$$x_1(t) = e^t + e^{-t} + e^{2t}, \qquad x_2(t) = e^t - e^{-t}.$$

9. Use Laplace transform to solve the initial value problem

$$\begin{cases} 2y'' + y' + 2y = \delta(t-5), \\ y(0) = y'(0) = 0, \end{cases}$$

where δ is the delta function.

Solution: Let $Y(s) = \mathcal{L}[y]$. Applying the Laplace transform leads to

$$(2s^2 + s + 2)Y(s) = e^{-5t}.$$

Thus

$$Y(s) = \frac{e^{-5s}}{2s^2 + s + 2} = \frac{e^{-5s}}{2} \frac{1}{\left(s + \frac{1}{4}\right)^2 + \frac{15}{16}}.$$

Recall that

$$\mathcal{L}^{-1}\left\{\frac{1}{\left(s + \frac{1}{4}\right)^2 + \frac{15}{16}}\right\} = \frac{4}{\sqrt{15}} e^{-t/4} \sin \frac{\sqrt{15}}{4} t.$$

Therefore

$$y(t) = \mathcal{L}^{-t}[Y(s)] = \frac{2}{\sqrt{15}} u_5(t) e^{-(t-5)/4} \sin\left(\frac{\sqrt{15}}{4}(t-5)\right).$$